

✓ Assignment – 12

🌐 “Smart Emergency & City Service Portal” (One Page Application)

🎯 Real-World Problem Statement

The Municipal Corporation wants to build a **Smart Emergency & Citizen Service Portal** that:

- Detects the user's current location
- Allows users to drag emergency services into priority area
- Stores user preferences locally
- Runs heavy background calculations without freezing the page
- Works as a single responsive web page

You are appointed as a Web Developer to design this **complete one-page Smart Portal** using today's HTML5 APIs.

📌 Important Rule

- Create **ONLY ONE HTML file**
 - Web Worker must be in **separate .js file**
 - Use proper layout and styling
 - Use real-world content
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🏠 Project Title:

`smart_city_emergency_portal.html`

Worker file:

`emergency_worker.js`

□ Functional Requirements

◆ 1 Page Layout (Semantic Structure Required)

Use:

- `<header>` – City Name & Logo
- `<nav>` – Navigation menu
- `<section>` – Main content
- `<article>` – Service details
- `<aside>` – Emergency contacts
- `<footer>` – Contact & copyright

Use Flexbox or Grid layout.

📍 2 Geolocation API Section

Create a section titled:

“Find My Current Location”

Requirements:

- Button → “Detect My Location”
- Use `navigator.geolocation.getCurrentPosition()`
- Display:
 - Latitude
 - Longitude
 - Accuracy
 - Timestamp
- Handle errors properly:
 - Permission denied
 - Position unavailable
 - Timeout
 - Unknown error

Optional:

- Embed Google Maps iframe using latitude & longitude

3 Drag and Drop Section

Create a section titled:

“Emergency Service Priority Board”

Scenario:

User can drag emergency services into “High Priority” area.

Draggable items:

-  Ambulance
-  Police
-  Fire Department
-  Hospital

Requirements:

- Use `draggable="true"`
- Use:
 - `ondragstart`
 - `ondragover`
 - `ondrop`
- Use `dataTransfer.setData()` and `getData()`
- Allow dropping inside a target `<div>`

Make it visually clear.

4 Web Storage Section

Create:

“Save My Preferences”

Add:

- Input field for User Name
- Select preferred emergency service

- Save button

Use:

localStorage:

- Store user name permanently
- Display saved name when page reloads

sessionStorage:

- Count how many times user clicked “Emergency Help” button
- Reset when tab closes

Also:

- Add remove button (`localStorage.removeItem()`)
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5 Web Worker Section

Create:

“Emergency Alert Counter”

Scenario:

City alert system runs heavy background calculations.

Requirements:

- Create `emergency_worker.js`
- Worker should:
 - Count numbers every 500ms
 - Use `postMessage()`
- In HTML:
 - Start Worker button
 - Stop Worker button
 - Display result in `<output>` element
- Use:
 - `new Worker()`
 - `onmessage`
 - `terminate()`

Important:
Worker must not freeze page.

6 Responsive Design

Include:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Ensure layout works on mobile.

7 Required Technical Checklist

Feature	Must Include
getCurrentPosition()	✓
Error handling (switch case)	✓
watchPosition() (optional bonus) ★	
Drag start handler	✓
Drag over handler	✓
Drop handler	✓
localStorage.setItem	✓
localStorage.getItem	✓
localStorage.removeItem	✓
sessionStorage counter	✓
Worker support check	✓
Worker start/stop	✓
postMessage()	✓

UI Suggestions

Your page should visually contain:

- Location panel
- Emergency drag board
- Saved user section
- Background worker counter
- Clear buttons
- Proper headings
- Clean layout

Submission Guidelines

- One HTML file
- One JS worker file
- Proper indentation
- Meaningful content
- No template copying

Evaluation Rubric (30 Marks Example)

Section	Marks
Layout & Structure	5
Geolocation Implementation	5
Drag & Drop	5
Web Storage	5
Web Worker	5
Code Quality & Responsiveness	5